We have seen that the power series $\sum_{k=0}^{\infty} x^k$ converges when $|x| < 1$, in which case $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$. Turning this around, we can view $\sum_{k=0}^{\infty} x^k$ as a representation of the function $f(x) = \frac{1}{1-x}$ defined on the interval $(-1,1)$. We will soon see that representing functions by power series is a useful strategy for integrating functions that don't have elementary antiderivatives, for solving differential equations, and for approximating functions by polynomials.

1. Express $\frac{1}{1+x^2}$ as a power series and find the interval of convergence.

2. Find a power series representation for $\frac{1}{x+2}$ and find the interval of convergence.

3. Find a power series representation for $\frac{x^3}{x+2}$ and find the interval of convergence.

We can also use the Theorem below, which we proved last time, to represent functions by power series.

Theorem. Suppose $\sum_{n=0}^{\infty} c_n(x-a)^n$ has radius of convergence $R > 0$. Then the function $f(x) = \sum_{k=0}^{\infty} c_n(x-a)^n$ is differentiable (and therefore continuous and therefore integrable) on the interval $(a-R, a+R)$ and the derivative (resp., antiderivative) is obtained by term-by-term differentiation (resp., integration)

(i) $f'(x) = \sum_{k=1}^{\infty} nc_n(x-a)^{n-1}$

(ii) $\int f(x)\,dx = C + \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}$

The radii of convergence of the power series in Equations (i) and (ii) are both R.

Remarks.

(a) Equations (i) and (ii) above can be rewritten as
   (i) $\frac{d}{dx}\left[\sum_{n=0}^{\infty} c_n(x-a)^n\right] = \sum_{n=0}^{\infty} \frac{d}{dx} c_n(x-a)^n$
   (ii) $\int \left[\sum_{n=0}^{\infty} c_n(x-a)^n\right] = \sum_{n=0}^{\infty} \int c_n(x-a)^n$

(b) While the theorem above says that the radius of convergence remains the same when a power series is differentiated or integrated, it does not say that the *interval* of convergence necessarily remains the same. It may happen, for instance, that the original series converges at an endpoint, whereas the differentiated series diverges there.

(c) The idea of differentiating a power series term by term is the basis for a powerful method for solving differential equations.

4. We have seen that the Bessel function

$$J_0(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2^{2n}(n!)^2}$$

is defined for all x (i.e., $R = \infty$). Thus, by the theorem above, J_0 is differentiable for all x and its derivative is found by term-by-term differentiation. Find $J_0'(x)$.

5. Express $\frac{1}{(1-x)^2}$ as a power series by differentiating $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$. What is the radius of convergence?

6. Find a power series representation of $\ln(1+x)$. *Hint*: $\frac{d}{dx}\ln(1+x) = \frac{1}{1+x}$.

7. Find a power series representation for $f(x) = \tan^{-1}(x)$. *Hint*: $\frac{d}{dx}\tan^{-1}(x) = \frac{1}{1+x^2}$.